

-- Cox Regression - Sarcopenia Onset -----

i Definition : ewgsop2
i Covariates : fixed
i Dairy : continuous
i Route : mice
i Interaction: none

-- MICE route - pooling across imputed datasets --

Detected 20 imputed datasets.
i Dairy modelled as natural cubic spline (ns, df = 3).
Fitting .imp = 1 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 49 events
Fitting .imp = 2 / 20 ...
Events: 57 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 51 events
Fitting .imp = 3 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 52 events
Fitting .imp = 6 / 20 ...
Events: 57 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 51 events
Fitting .imp = 7 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 50 events
Fitting .imp = 8 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 49 events
Fitting .imp = 9 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 50 events
Fitting .imp = 10 / 20 ...
Events: 58 sarcopenia cases out of 1121 participants

??? Fixed dataset: 1121 rows | 50 events
Fitting .imp = 12 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 50 events
Fitting .imp = 13 / 20 ...
Events: 57 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 51 events
Fitting .imp = 14 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 50 events
Fitting .imp = 15 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 1121 rows | 50 events
Fitting .imp = 16 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants

```
??? Fixed dataset: 1121 rows | 51 events
Fitting .imp = 18 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants
```

```
-- Fixed covariates dataset (one row per pt) --
```

```
??? Fixed dataset: 1121 rows | 49 events
Fitting .imp = 19 / 20 ...
Events: 55 sarcopenia cases out of 1121 participants
```

```
-- Fixed covariates dataset (one row per pt) --
```

```
??? Fixed dataset: 1121 rows | 49 events
Fitting .imp = 20 / 20 ...
Events: 56 sarcopenia cases out of 1121 participants
```

```
-- Fixed covariates dataset (one row per pt) --
```

```
??? Fixed dataset: 1121 rows | 50 events
```

```
-- Pooled adjusted model results (Rubin's rules) --
```

```
# A tibble: 10 x 5 term HR CI_low CI_high p.value <chr> <dbl> <dbl> <dbl> <dbl>
1 splines::ns(dairy_exposure, df = 3)1 4.82 0.377 61.7 2.19e-1 2
splines::ns(dairy_exposure, df = 3)2 0.0172 0.0000419 7.08 1.80e-1 3
splines::ns(dairy_exposure, df = 3)3 0.000988 0.0000000256 38.1 1.92e-1 4 BMI
0.690 0.614 0.775 1.61e-7 5 education_levelMedium (ISCED 3-4) 1.03 0.450 2.38
9.34e-1 6 education_levelHigh (ISCED 5-8) 1.03 0.327 3.25 9.58e-1 7
smoking_statusFormer 0.886 0.465 1.69 7.07e-1 8 smoking_statusCurrent 0.821
0.307 2.19 6.86e-1 9 mvpa_min_day_f1 0.999 0.996 1.00 5.18e-1 10
```

```
-- Assumption checks [mice_imp1] --
```

```
-- 1. Proportional Hazards - cox.zph() | pass: global p > 0.05
```

```
chisq df p splines::ns(dairy_exposure, df = 3) 3.88135 3 0.275 BMI 0.00989 1
0.921 education_level 5.22930 2 0.073 smoking_status 4.77321 2 0.092
mvpa_min_day_f1 0.46461 1 0.495 diabetes_status 1.66510 1 0.197 GLOBAL 16.42589
10 0.088
??? PH assumption not violated (global p = 0.0881)
Schoenfeld plots saved to
03_outputs/Cox/mice_ewgsop2_fixed_continuous_spline3_27072026_1238
```

```
-- 2. Linearity - martingale (null model) vs continuous predictors | pass: smooth ??? linear
```

```
'geom_smooth()' using formula = 'y ~ x'
'geom_smooth()' using formula = 'y ~ x'
Linearity plots saved to
03_outputs/Cox/mice_ewgsop2_fixed_continuous_spline3_27072026_1238
```

```
-- 3. Outliers - deviance & martingale residuals | pass: |dev| <= 3
```

```
i 2 observation(s) with |deviance residual| > 3.
```

```
-- 4. Influential observations - std. DFBetas | pass: no extreme spikes
```

```
i 25 obs exceed |DFBeta_std| > 0.2.
```

```
-- 5. Overall Fit - Cox-Snell residuals | pass: points on 45° diagonal
```

```
Cox-Snell plot saved to
03_outputs/Cox/mice_ewgsop2_fixed_continuous_spline3_27072026_1238
```

```
-- 6. Multicollinearity - VIF | pass: VIF < 5 (< 10 relaxed)
```

```
GVTE Df GVTE^(1/(2*Df)) splines::ns(dairy_exposure, df = 3) 1.118 3 1.019 BMI
```

-- Linearity test: spline vs linear dairy term

$\chi^2(2) = 2.87$, $p = 0.239$

No significant departure from linearity ($p \geq 0.05$).

-- Incidence rates by dairy quartile [mice]

quartile	n_pts	n_events	person_years	rate_per_1000py	ci_low	ci_high
1 Q1 (low)	281	10.00	19904.70	0.50	0.24	0.92
2 Q2	280	9.70	19833.92	0.49	0.23	0.91
3 Q3	280	13.00	19898.40	0.65	0.35	1.12
4 Q4 (high)	280	17.45	19946.00	0.87	0.51	1.39

Incidence table saved to

03_outputs/Cox/mice_ewgsop2_fixed_continuous_spline3_27072026_1238

Model results saved to

03_outputs/Cox/mice_ewgsop2_fixed_continuous_spline3_27072026_1238/mice_model_results.csv